

# MAT627 Chapter 2.4 Programming Assignment

Emre Tekmen

April 20, 2026

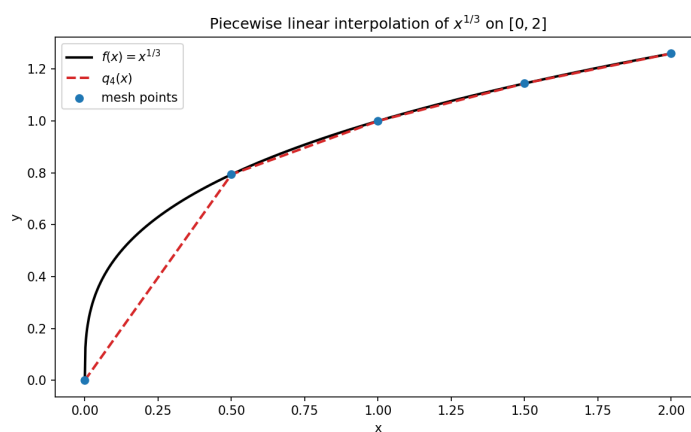


Figure 1: Part 1:  $q_4(x)$  and  $f(x) = x^{1/3}$  on  $[0, 2]$ .

## 1 Part 2

Uniform piecewise linear interpolation on  $[1, 2]$ .

## 2 Part 3

Uniform piecewise linear interpolation on  $[0, 1]$ .

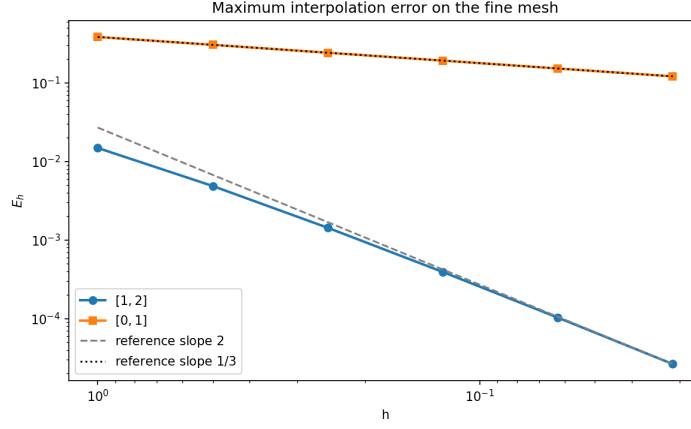


Figure 2: Error decay on the fine grid for the intervals  $[1, 2]$  and  $[0, 1]$ .

Table 1: Maximum fine-grid error for  $f(x) = x^{1/3}$  on  $[1, 2]$ .

| $N$ | $h$     | $E_h = \max_{z_i \in D}  q_N(z_i) - f(z_i) $ | Rate     |
|-----|---------|----------------------------------------------|----------|
| 1   | 1       | $1.488703 \times 10^{-2}$                    | —        |
| 2   | 0.5     | $4.875503 \times 10^{-3}$                    | 1.610433 |
| 4   | 0.25    | $1.434610 \times 10^{-3}$                    | 1.764892 |
| 8   | 0.125   | $3.929018 \times 10^{-4}$                    | 1.868418 |
| 16  | 0.0625  | $1.031247 \times 10^{-4}$                    | 1.929779 |
| 32  | 0.03125 | $2.641642 \times 10^{-5}$                    | 1.964883 |

Table 2: Maximum fine-grid error for  $f(x) = x^{1/3}$  on  $[0, 1]$ .

| $N$ | $h$     | $E_h = \max_{z_i \in D}  q_N(z_i) - f(z_i) $ | Rate     |
|-----|---------|----------------------------------------------|----------|
| 1   | 1       | $3.848998 \times 10^{-1}$                    | —        |
| 2   | 0.5     | $3.054952 \times 10^{-1}$                    | 0.333333 |
| 4   | 0.25    | $2.424717 \times 10^{-1}$                    | 0.333333 |
| 8   | 0.125   | $1.924499 \times 10^{-1}$                    | 0.333333 |
| 16  | 0.0625  | $1.527476 \times 10^{-1}$                    | 0.333333 |
| 32  | 0.03125 | $1.212358 \times 10^{-1}$                    | 0.333333 |

### 3 Discussion

On  $[1, 2]$ , the second derivative

$$f''(x) = -\frac{2}{9x^{5/3}}$$

is bounded, so the standard piecewise linear interpolation estimate applies and the observed rate is approximately 2.

On  $[0, 1]$ , the second derivative is singular at  $x = 0$ , so the theorem does not apply. On the first cell  $[0, h]$ , writing  $x = th$  with  $0 \leq t \leq 1$ , the interpolant is

$$q_N(x) = th^{1/3},$$

which gives

$$|f(x) - q_N(x)| = h^{1/3}|t^{1/3} - t|.$$

Therefore the dominant error behaves like  $O(h^{1/3})$ , so the rate is not second order near the origin and instead approaches  $1/3$ .